

Research Plan

- Artificial Intelligence-Aided Tour Design and Trajectory Optimization

Recent advances in artificial intelligence (AI) are reshaping aerospace engineering. In astrodynamics, AI capability is growing rapidly, yet trajectory design remains mission-specific and bottlenecked by expensive nonlinear solvers, the combinatorial growth of the trade space, and sensitivity in multi-body and chaotic dynamics. These constraints often force simplifying assumptions and limit rapid trade studies. At the same time, future missions increasingly require onboard autonomy, including contingency-aware replanning and guidance-level decision-making under uncertainty, which motivates fast, robust trajectory-design tools. To address these challenges, I leverage my expertise in **AI-aided tour design (multi-flyby sequencing) and trajectory optimization** across impulsive, low-thrust, and multi-body regimes, with applications to interplanetary, cislunar, and moon-tour missions. Figure 1 summarizes how my work connects AI methods to tour design through core algorithmic modules, research goals, and target applications.

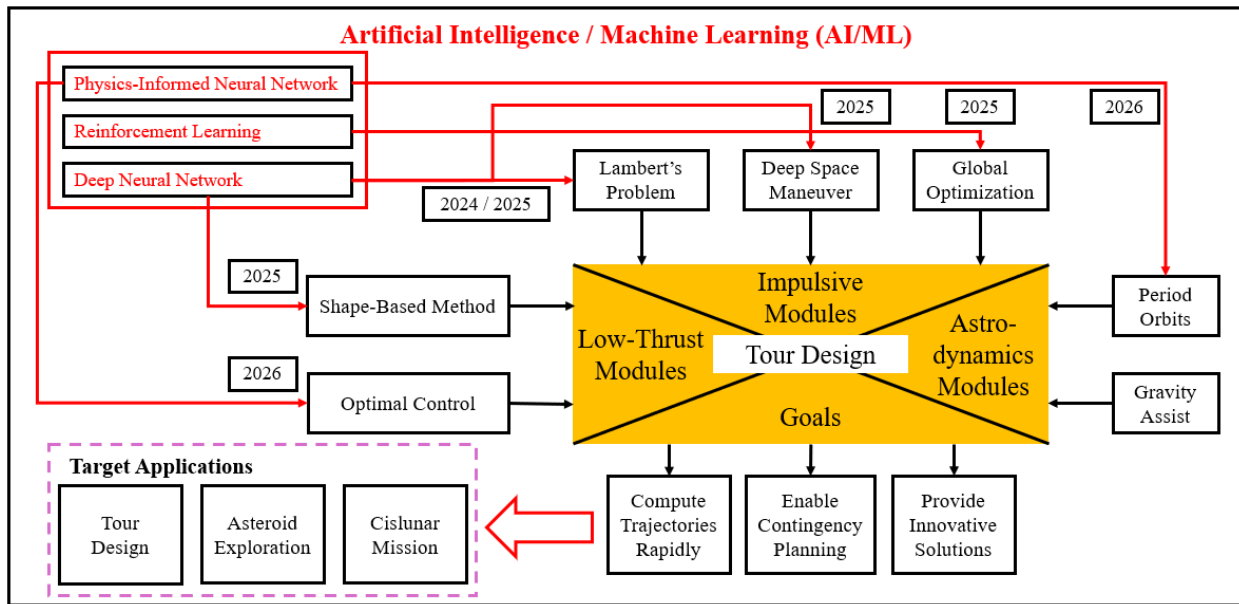


Figure 1. Overview of my past and current AI-aided tour design and trajectory optimization research and its target applications.

- JPL: Impulsive & Continuous-thrust Tour Design

During my internship at JPL, I learned that robust trajectory design is increasingly important in practice: operational anomalies and modeling uncertainties can drive a spacecraft away from its nominal trajectory. Motivated by this challenge, my doctoral research advanced impulsive tour design by combining neural networks (NNs) and optimization. I developed an **NN Lambert solver** achieving up to a **7× speedup** over a conventional approach, and a **vectorized, dynamic-size genetic algorithm** that optimizes four-flyby sequences in about two minutes without parallelization. I also introduced a **reinforcement-learning (RL)** approach that learns an approximate cost-to-go (estimated remaining mission cost) for transfer decisions, reducing the need for repeated full-trajectory optimization during

sequential planning. In parallel, I proposed **NN-based deep-space maneuver (DSM) prediction** to reduce search dimensionality by eliminating the need to solve a new optimization problem at each transfer. Moving forward, I will leverage these modules to build a contingency-aware framework in close collaboration with JPL. In such scenarios, RL policies can provide rapid recommendations from the current state when off-nominal events occur, enabling rapid replanning without full re-optimization. I will further integrate DSM prediction into multi-gravity-assist tours and expand the objective beyond nominal mission cost to explicitly account for uncertainty and recovery capability.

In addition to impulsive trajectories, I will pursue collaborations with JPL on continuous-thrust trajectory optimization. Continuous-thrust trajectory design, such as **solar electric propulsion (SEP)** and **solar-sail** transfers, is challenging to solve reliably, and it becomes even more difficult in operations when duty-cycle constraints and off-nominal conditions need to be modeled. My prior work on NN-based mission-cost prediction with shape-based methods reduces global-search time **from tens of minutes to seconds**, narrowing the search region for high-fidelity optimization. I am currently investigating Extreme Theory of Functional Connections (**X-TFC**), a physics-informed neural net that has shown strong performance on two-point boundary-value problems as a scalable solver for continuous-thrust trajectories. I plan to present results on applying this framework to SEP at an upcoming Astrodynamics Specialist Conference and to extend it to solar-sail transfers. Ultimately, I aim to connect these advances to **Q-law**, a foundational algorithm for continuous-thrust transfers at JPL, using X-TFC as a physics-constrained solver and integrating **RL** to support robust design under uncertainty.

- Lockheed Martin & NSF: Moon-Tour Problem

Moon-tour trajectory design has demonstrated industry relevance (e.g., published studies supported by Lockheed Martin) and motivates fundamental advances in mathematical methods aligned with NSF priorities. A moon tour seeks a sequence of moon flybys that reshapes the spacecraft's orbit under constraints with minimal propellant. However, the design space grows combinatorially with the number of moons, resonances, and flyby sequences, making rapid trade-space exploration computationally intractable. Building on my JPL experience in moon-tour design, where V_{∞} -leveraging transfers (VILTs) and resonance-hopping are used, I will develop tools that move beyond standard assumptions in impulsive maneuvers (e.g., a single tangential impulse at periapsis/apoapsis). To relax these assumptions, I will apply **primer-vector theory** and solve the optimality conditions using **X-TFC**. I will then extend this framework to **low-thrust VILTs** by solving the associated optimal-control problems. Finally, to enable end-to-end tour construction, I will explore graph neural networks (GNNs) that represent resonance transitions explicitly. In this representation, resonance ratios define nodes and feasible transfers define edges, enabling efficient selection of resonance and flyby sequences. Together, these efforts will deliver an end-to-end moon-tour design pipeline that makes the vast, combinatorial trade space computationally tractable.

- NASA & AFOSR: Autonomy in Cislunar Space

NASA and Air Force Office of Scientific Research (AFOSR) have growing interest in cislunar missions. Cislunar operations are more accessible to small spacecraft, yet limited communication, sparse tracking, and the need to coordinate multiple spacecraft demand greater onboard autonomy. In this thrust, I will develop autonomous multi-spacecraft trajectory optimization methods in cislunar space using AI/ML.

I will first use **Gaussian processes (GPs)** to model smooth trajectories and generate high-quality training data that preserves temporal correlation and uncertainty structure. While GPs represent smooth trajectories well, they are typically conditioned on an initial state, making continuation from an intermediate state challenging. To address this, I will train a **Transformer-based sequence model** on GP-generated trajectories so that, given an intermediate state and mission context, the model can predict the remaining trajectory and the corresponding guidance commands (control sequence). This GP-to-Transformer pipeline will enable rapid trajectory continuation and replanning for autonomous multi-spacecraft operations in cislunar space.

- JSC, SNL, Holloman AFB, DoW: Learning-Augmented Space Navigation

Space navigation is a core component of guidance, navigation, and control (GNC), and recent missions increasingly require accurate position and attitude determination. My research includes spacecraft navigation, which aligns well with the interests of NASA Johnson Space Center (JSC), Sandia National Laboratories (SNL), Holloman Air Force Base (Holloman AFB), and the United States Department of War (DoW). In a related project, I formulated coupled spacecraft position and attitude determination as a geometry-based root-finding problem using angles-only measurements. I am extending this framework to initial orbit determination and recursive state estimation using an Extended Kalman Filter. In future work, I will augment model-based recursive filtering with **Long Short-Term Memory (LSTM)** networks to learn time-correlated residual patterns from sequential measurements and improve estimation performance in realistic conditions. I will also investigate an **encoder–decoder** model to detect anomalies and unmodeled maneuvers, enabling more resilient navigation for space missions. This intelligent, learning-augmented navigation will support reliable GNC performance for future missions that operate with limited tracking and complex operational environments.

- New Mexico Space Research Ecosystem

New Mexico hosts a unique national-security and space research ecosystem that includes White Sands Missile Range (WSMR), Kirtland Air Force Base (Kirtland AFB), Los Alamos National Laboratory (LANL), and Sandia National Laboratories (SNL). I will pursue collaborations across this ecosystem by connecting my expertise in AI/ML-enabled trajectory design and autonomy to applications aligned with their research interests, particularly **space situational awareness (SSA)** and **atmospheric entry**. First, to collaborate with **WSMR**, I will extend my research toward SSA, leveraging WSMR's radar-based tracking capabilities, including the Deep-space Advanced Radar Capability (DARC). With **LANL**, I will advance AI/ML methods for scalable orbit prediction, anomaly detection, and uncertainty-aware risk assessment. At **SNL**, where Trajectory Analysis and Optimization Software (TAOS) supports atmospheric re-entry analysis, I will broaden my expertise to re-entry applications in addition to my interplanetary mission design work and pursue collaborative extensions of TAOS. I will also pursue collaboration with **Kirtland AFB**, where research on atmospheric drag and density modeling can strengthen re-entry trajectory optimization. Together, these collaborations will position NMSU to deliver an end-to-end astrodynamics pipeline that connects sensing and estimation to decision-making, including conjunction screening, maneuver planning, and entry guidance, with clear pathways to on-orbit servicing, collision avoidance, and debris mitigation.

- Internal Research Collaboration

New Mexico State University's emphasis on cross-disciplinary research aligns strongly with my vision of AI/ML-enabled multidisciplinary research for autonomous systems, integrating optimization, estimation, and control across diverse dynamical platforms. Within NMSU, I see natural synergies with Prof. Luis Guerrero-Bonilla on autonomy and resiliency in **multi-robot systems**, where my background in PINNs and RL can contribute to optimal path planning and decision-making. I also anticipate strong collaboration with Prof. Mahdi Haghshenas-Jaryani on bio-inspired robotics, where my AI/ML background can support planning and learning-enabled control for hybrid robotic systems. These methods can be evaluated in hardware-focused testbeds, including agricultural robotics platforms. In addition, we can explore a new direction in **bio-inspired machine learning** (biologically motivated learning rules and architectures) as a learning-side counterpart to bio-inspired optimization methods such as genetic algorithms and ant-colony optimization. Finally, I see complementary opportunities with Prof. Francisco Torres Herrador on **aerobraking**. In prior work, I studied multidisciplinary coupled optimization that integrates trajectory design with spacecraft design for Mars aerobraking. Through collaboration, I will extend this foundation by incorporating higher-fidelity atmospheric models and coupling aerobraking trajectories with thermal and structural design constraints.

- AIM Lab (proposed): Research Vision and Funding Alignment

I plan to establish the AeroAstro Intelligent Mechanics (AIM) Lab to advance AI-aided trajectory design with an emphasis on robust, uncertainty-aware methods, contingency planning, and scalable computation. In the first two years, I will establish the core methodology of X-TFC for moon-tour design through (1) low-thrust trajectory optimization and (2) primer-vector-guided optimality analysis, and I will leverage these foundations to develop RL-based contingency planning. In parallel, I will extend my navigation research by integrating LSTM with filtering to improve robustness under real-world uncertainties. These efforts will produce open-source software, benchmark cases and datasets, and early journal and conference publications, forming the foundation for competitive external proposals to the agencies identified above.

While the thrusts above focus on **AI-aided mission design**, AIM's research agenda will also include AI-driven **spacecraft navigation** and **SSA**. More broadly, AIM will pursue AI/ML-enabled multidisciplinary research that integrates modeling, optimization, and machine learning with applications across multiple domains. Methodologically, AIM will balance physics-based **dynamical systems** and **optimization** with **machine learning**, advancing both applications and fundamental techniques. I will translate these directions into competitive proposals and pursue external support from NASA and NSF and, where appropriate, AFOSR and DoW. I will continue building collaborations with JPL and cultivate new academic partnerships to create impactful projects and strong training opportunities for students. In doing so, AIM will reduce reliance on simplifying assumptions, enable rapid trade studies, and support contingency-aware autonomy for future missions.